Lab:5

Title :FM modulation and Demodulation

OBJECTIVE: to become familiar with FM modulation and demodulation

Material required:

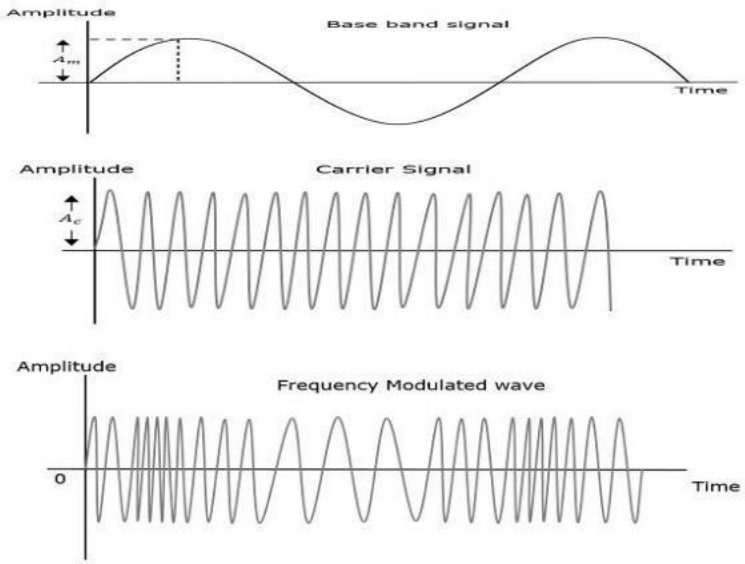
THEORY:

When the frequency of the carrier wave varies with the amplitude of the message signal, it is called **frequency modulation. (Type of angle modulation)** The carrier signal is sent with the message signal during the modulation process. A message signal is also known as the **baseband signal**. It is nothing but a signal containing information that needs to be transmitted from the **sender** to the **receiver**.

A sender in the communication system is known as a **transmitter**.

Angle Modulation is the process in which the frequency or the phase of the carrier signal varies according to the message signal.

In amplitude modulation, the amplitude of the carrier signal varies. Whereas, in Frequency Modulation (FM), the frequency of the carrier signal varies in accordance with the instantaneous amplitude of the modulating signal. Hence, in frequency modulation, the amplitude and the phase of the carrier signal remains constant.



The frequency of the modulated wave increases, when the amplitude of the modulating or message signal increases. Similarly, the frequency of the modulated wave decreases, when the amplitude of the modulating signal decreases.

The equation for instantaneous frequency fi in FM modulation is fi=fc+kfm(t) Where, fc is the carrier frequency kt is the frequency sensitivity m(t) is the message signal.

If the modulating signal is $m(t) = A_m \cos(2\pi f_m t)$, then the equation of FM wave will be

$$s(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$$

Where,

$$\beta = \textit{modulation index} = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$$

Frequency modulation (FM) is the encoding of information in a carrier wave by varying the instantaneous frequency of the wave. The amount of frequency variation is proportional to the Modulation index.

A major advantage of FM as compared with AM, is improved Signal-to-noise ratio (SNR).

PROGRAM:

```
clc;
clear all;
close all;
%sets the parameters
fs=10000;
t=(0:0.1*fs)/fs;
Am=input('enetr the amplitude of message signal=');
Ac=input('enter the amplitude of carrier signal=');
Fm=input('enter the frequency of message signal=');
Fc=input('enter the frequency of carrier signal=');
m = input('Enter Modulation Index = ');
%equation of message signal
Mt=Am*cos(2*pi*Fm*t);
subplot(4,1,1);
plot(t,Mt);
xlabel('time');
ylabel('amplitude');
title('message signal');
%equation of carrier signal
Ct=Ac*cos(2*pi*Fc*t);
subplot(4,1,2);
plot(t,Ct);
xlabel('time');
ylabel('amplitude');
title('carrier signal');
%equation of modulated signal
St=Ac*cos((2*pi*Fc*t)+m*sin(2*pi*Fm*t));
subplot(4,1,3);
plot(t,St);
xlabel('time')
ylabel('amplitude');
title('modulated signal');
%demodulated signal
d=demod(St,Fc,fs,'Fm');
subplot(4,1,4);
plot(t,d);
xlabel('time');
```

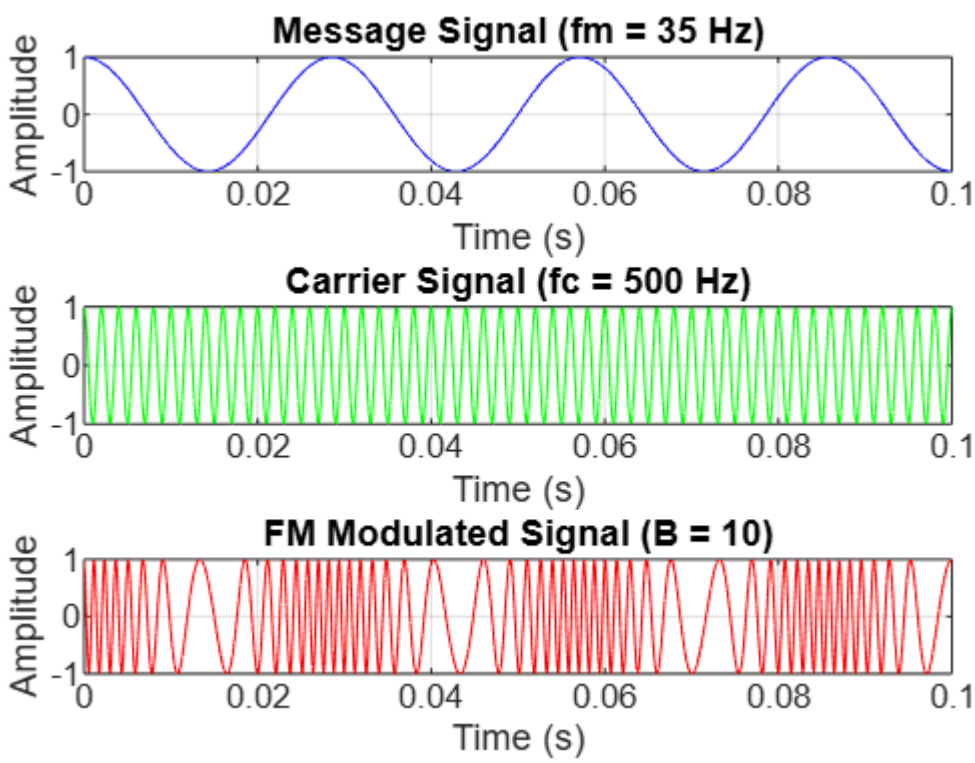
```
ylabel('amplitude');
title('demodulated signal');
```

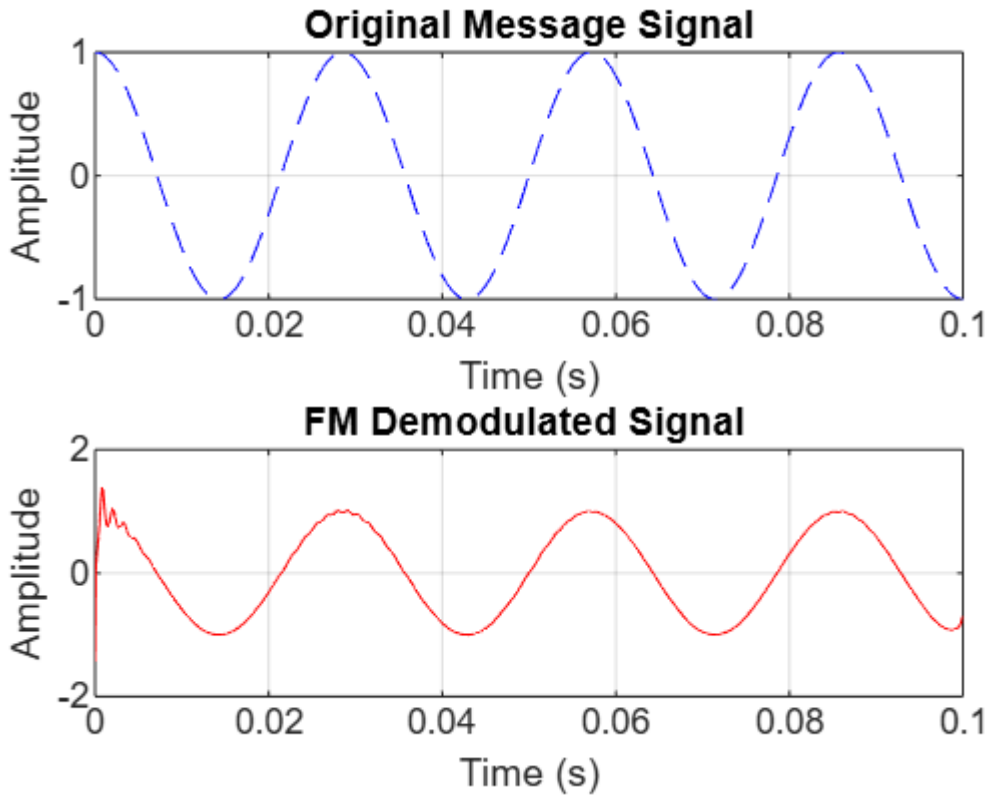
OUTPUT:
Am=amplitude of message signal:1
AC=amplitude of carrier signal:1
Fm=frequency of message signal:35
Fc=Frequency of carrier signal:500
M=modulation index=10.

%**********second way**********/

```
clc;
clear;
close all;

% Parameters
fs = 10000;          % Sampling frequency (Hz)
Am = 1;              % Message amplitude
Ac = 1;              % Carrier amplitude
fm = 35;             % Message frequency (Hz)
fc = 500;            % Carrier frequency (Hz)
B = 10;              % Modulation index
freqdev = B * fm;    % Frequency deviation (Δf = B * fm)
kf = 2 * pi * freqdev;  % Frequency sensitivity

t = (0:0.1*fs)/fs;   % Time vector (0 to 0.1 seconds)

% Message signal
m_t = Am * cos(2 * pi * fm * t);

% Carrier signal (for reference)
c_t = Ac * cos(2 * pi * fc * t);

% === FM Modulation (manual) ===
int_m = cumsum(m_t) / fs;                    % Integral of message
s_t = Ac * cos(2*pi*fc*t + kf * int_m);      % FM signal

%% ---------------- Figure 1: Modulation ----------------
figure(1);

subplot(3,1,1);
plot(t, m_t, 'b');
title('Message Signal (fm = 35 Hz)');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(t, c_t, 'g');
title('Carrier Signal (fc = 500 Hz)');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;

subplot(3,1,3);
plot(t, s_t, 'r');
title('FM Modulated Signal (B = 10)');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;

%% ========== FM Demodulation (manual, using phase) ==========
```

```
% Hilbert Transform to get analytic signal
s_analytic = hilbert(s_t);                      % Complex envelope
inst_phase = unwrap(angle(s_analytic));         % Instantaneous phase
inst_freq = [0 diff(inst_phase)] * fs / (2*pi);  % Instantaneous freq (Hz)

% Remove carrier frequency to get message
m_demod = (inst_freq - fc) / freqdev;           % Normalized back to message

%% ---------------- Figure 2: Demodulation ----------------
figure(2);

subplot(2,1,1);
plot(t, m_t, 'b--');
title('Original Message Signal');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;

subplot(2,1,2);
plot(t, m_demod, 'r');
title('FM Demodulated Signal ');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;
```





Discussion and conclusion: